Deep Copy

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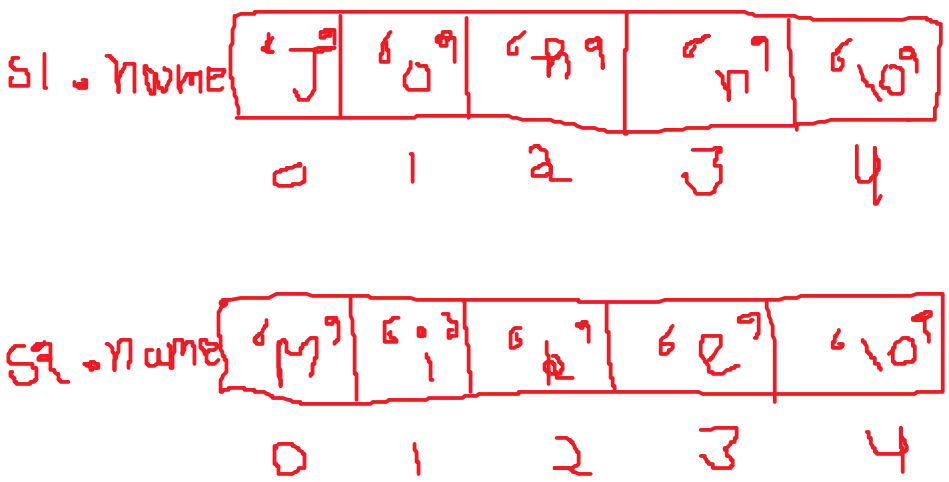
Deep Copy

- Means copying the actual content pointed by the pointer.
- Ensures each object has its own independent copy of the data.

```
#include<iostream>
#include<cstring>

class Student
{
        char *name;
    public:
        Student (const char *n)
        {
            name = new char[strlen (n) + 1];
            strcpy (name, n);
        }
        Student (const Student &s)
        {
            name = new char[strlen (s.name) + 1];
            Strcpy (name, s.name);
        }
        void setName (const char *n)
        {
            strcpy (name, n);
        }
        void display ()
        {
            std :: cout << "Name: " << name << "\n";
        }
        ~Student ()
        {
            delete[] name;
        }
};

int main ()
{
    Student s1 ("John");
    Student s2 = s1;     // deep copy
    s2.setName ("Mike");
    s1.display ();
    s2.display ();
    return 0;
}
```



Output:
Name: John
Name: Mike